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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.6

$$\int \frac{9}{4x^3-3x+1}dx$$

Probeer $\frac{1}{1,2,4}$, nulpunt bij $x=\frac{1}{2}$

$$\int \frac{9}{(2x-1)(2x^2+x-1)}dx$$

$$\Delta = 1^2 - 4\cdot 2\cdot(-1) = 9 \qquad x_{1,2} = \frac{-1\pm\sqrt{9}}{2\cdot 2} \sim x_1 = \frac{1}{2},\ x_2 = -1$$

$$\int \frac{9}{(2x-1)^2(x+1)}dx = \frac{A}{2x-1} + \frac{B}{(2x-1)^2} + \frac{C}{x+1}$$

Dan is

$$B = \left.\frac{9}{x+1}\right|_{x=\frac{1}{2}} = 6$$

$$C = \left.\frac{9}{(2x-1)^2}\right|_{x=-1} = 1$$

$$\begin{cases} \frac{9}{(2x-1)^2(x+1)} - \frac{B}{(2x-1)^2} = \frac{9}{(2x-1)^2(x+1)} - \frac{6}{(2x-1)^2} \\ = \frac{9-6(x+1)}{(2x-1)^2(x+1)} = \frac{-6x+3}{(2x-1)^2(x+1)} \\ = \frac{-3(2x-1)}{(2x-1)^2(x+1)} = \frac{-3}{(2x-1)(x+1)} \end{cases}$$

$$A = \left.\frac{-3}{x+1}\right|_{x=\frac{1}{2}} = -2$$

$$\Rightarrow \int \left(\frac{-2}{2x-1} + \frac{6}{(2x-1)^2} + \frac{1}{x+1}\right)dx$$

$$= \int \frac{-2}{2x-1}dx + \int \frac{6}{(2x-1)^2}dx + \int \frac{1}{x+1}dx$$

$$t = 2x-1 \left(dt = 2dx, dx = \frac{dt}{2}\right)$$

$$= \int \frac{-2}{2x-1}dx + 6\int t^{-2}\frac{dt}{2} + \int \frac{1}{x+1}dx$$

$$= \int \frac{-2}{2x-1}dx + 3\int t^{-2}dt + \int \frac{1}{x+1}dx$$

$$= -\ln|2x-1| - \frac{3}{2x-1} + \ln|x+1| + C$$

References